

Controlling air traffic

A new technology called **FACES** (Free flight Autonomous and Coordinated Embarked Solver) allows planes to swarm the air and can prevent mid air collisions, regardless of the traffic. FACES is being developed by the Centre For Aerial Navigation Studies, France and lets pilots choose their own routes immediately after take off, instead of forcing them to fly a predetermined path. The software advises pilots to make manoeuvres only when absolutely necessary and maps out the best routes, saving on travel time.



The theory of distributed software is applied so no particular plane is ever in control, and they make their avoidance manoeuvres together. FACES has already been tested on a simulator using archived data on real air traffic and it successfully prevented a pile-up with 35 planes sharing airspace. The software sets the plan, keeping in account the flight plans of other planes that have already set it into sequence. Once the routes have been set, the pilots carry out the choreographed manoeuvres. This will result in the non-colliding routes.

snapshot

Wireless networking industry expected to bring in \$2 billion revenue this year, and estimated to grow by 30%

Source: 802.11 Planet Conference

Archive your life!

In a curious twist of computing, Microsoft is creating a digital archive project where you can build a giant database of your life using the modern PC. Called **MyLifeBits**, the project aims to build archives that chronicle the events in people's lives. Add in all the photographs, home videos, audio recordings of conversations, letters and other official and unofficial documentation, and pretty soon, you should be able to run the equivalent of a Google search for any event in your



life. With data storage costs expected to come down massively, this could be feasible—calculate some 1,000 GB for 4 hours of videotape recording for everyday of your life.

This digital archive could be a boon for the archaeologists of a future generation trying to uncover our civilisation. While the security risks are pretty clear, an accurate, searchable store of events could help us preserve our experiences more vividly for posterity.

Internet buzz

Pinging a data packet to a remote computer and calculating the time taken to receive its echo is used to ascertain the reliability of the Internet connection. However, Chis Chafe, cellist and researcher at Stanford University, has modelled the Internet connection as guitar strings, twanging them to reveal those subtle characteristics that are

missed by ping.

The time taken by the pulse to bounce back depends on the state of the network. The longer the connection time, the lower was the frequency range reproduced. This works such that any sudden loss of sound indicates a broken link somewhere or missing packets of data, thus revealing the jitter.

quoteworthy

"Companies throughout the country have networks that are wide open because of wireless LANs... millions of houses are getting connected, which means that more and more are getting vulnerable"

Richard Clarke, special advisor to the US President for cyber security

"Everyone can hear what you're talking about on the phone. When you're IM-ing, everyone can hear you typing, but it sounds like you're working."

Melanie Miller, staff assistant at the Massachusetts Institute of Technology, on

tomorrow's technology

Molecular photography

Researchers at the Oklahoma University in the US have invented a method to store digital information on hydrogen atoms. The inventors of this technology, Bing Fung and his colleagues, found that the 19 hydrogen atoms in a single liquid crystal molecule can be used to store 1,024 bits of digital information. The information is stored in the interaction of the magnetic moments of the protons

within the atoms and is related to the spins of the subatomic particles of the atoms. In initial experiments, they succeeded in encoding a 32 pixel square monochrome image in a molecule where the information was stored in a binary format. This was done by firing a set of 1,024 different radio frequencies at the molecule. Each of the frequencies either had an amplitude (representing a binary 1) or did not (representing a 0).

The information thus stored was read by firing a second pulse at the molecule having a slightly different frequency. Consequently, the information was read by measuring the changes in the interference patterns using a Nuclear Magnetic Resonance (NMR) instrument.

While this technology is still a while away from commercial viability, it promises big things for storage.